Evaluation of Various Avatar Designs for Conversational Chatbot Systems

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Abstract

This paper examines multiple avatar designs applied to a state-of-the-art conversational chatbot system, and aims to determine how differences in gender, realism and humanity presented by the chatbot affects the user experience. For our experiment, we used a 20 turn conversation between a participant and our text only chatbot, then created 10 video simulations of the conversation each adopting a unique avatar. 6 participants then observed the simulations, and conducted evaluations by ranking the setups and rating their impressions. The results support that state-of-the-art conversational chatbots do indeed benefit from adopting avatars, however too much realism and human-ness could be detrimental - furthermore, participants expressed better opinion on avatars of the gender opposite to theirs, suggesting that gender is another key factor.

Key Words
Conversational Agents, Chatbot, Avatar, Communicative Behaviors

1. Introduction

Recently, with the development of language models such as GPT-3 by OpenAI (Brown et al., 2020) [1] which rely on using many billions of parameters for powerful performance, conversational chatbots have been able to leverage those developments to perform better than ever. One example of a successful conversational chatbot which relies on the GPT-3 model is Emerson AI by Quickchat. This chatbot is considered among the best out there for its ability to appear intelligent and hold conversations which do not feel overly artificial, which had been the case with most conversational chatbots in the past. The key reason for this success is the GPT-3 model's ability to understand language on a very deep level, allowing it to catch meanings in something as difficult as the user's tone and nuances. This is thanks to the extremely large amount of trainable parameters and data this language model used.

Much research has been conducted on avatars of systems, however this is not the case for the setting of conversational chatbots with the aim of carrying satisfying conversations with its users - especially with the previously mentioned developments in conversational chatbots, there has not been any studies considering its performance when adopting avatars. Thus, in our experiment, we set up 10 different avatar configurations on a state-of-the-art chatbot system to evaluate user experience with them. In regards to the differences between the avatars, we will have them vary in realism, humanity and gender, which have been determined to be significant factors in some previous works. The evaluations will be conducted using scores across some variables as well as rankings, designed overall to assess the ability of the chatbot to carry a satisfying conversation with the user.

2. Overview of avatars

As mentioned in the previous section, we have designed 10 different avatar configurations for the system to perform alongside a state-of-the-art conversational chatbot. The 10 different designs vary in realism, humanity and gender - in the following sections we will go over these in detail.

2.1 Moving images

For the most realistic set of avatars, we used moving photographic images. In this configuration we used a tool, MakeItTalk (Zhou et al., 2021) [2] to animate realistic facial images to match the system dialogue.

Regarding the images that we selected, we chose photographic images of a woman, a man and a robot. The reasoning behind selecting these three types of avatars is first to differentiate and study the effects of gender on the participants' experience. A 2020 study reports that most voice based conversational agents are designed to appear female without any significant reasoning other than the fact that most developers are male and have more positive impressions of women than men [3]. Thus, by choosing three variations which are male, female and undefined in terms of gender, we hoped to see if there would be any difference in how evaluators rated them. Below in Figure 1 are the three image avatars we have used.



Figure 1. Female, male and robot avatars animated using MakeItTalk

2.2 Animated cartoons

For this configuration, we used cartoon images of a man, a woman and a robot. Unlike the previous method, instead of animating the images using the tool MakeItTalk [1], we had the mouths of the cartoon images cycle through frames of different mouth positions to match the system dialogue. This was done in order to distinguish this configuration's lack of realism compared to the previous method. As MakeItTalk animates the images very similarly to how real human faces move when speaking, going as far as to making the avatars blink on occasion, using cartoon animations limited to rigid mouth movements was effective for our purpose.

The reason why we believed realism would be a significant factor is based off of a Penn State study which found that too much anthropomorphism in a chatbot avatar is actually detrimental to the user's experience with them [4]. This was due to the fact that if the avatar has more human-like cues in its design, the users expect intelligence to match the humanity - however, since conversational chatbots are not yet on the same level as real people, this causes greater disappointment where the chatbot falls short. Below in Figure 2, we have shown the three cartoon avatars that we used.



Figure 2. Cartoon female, male and robot avatars animated by cycling mouth animations across frames

2.3 Low resolution pixel art

For the next set of configurations, we used low resolution images in the style of pixel art in which the individual pixels of the image are visible as squares. Like previous configurations, we used three different designs: male, female and robot. As with the cartoon avatars, we aimed to have this configuration appear less realistic, however we went even further with this. In this design, the resolution was brought so low that we could not even animate the mouths to match the dialogue, as the mouths make up only two pixels.

In choosing this design, we again addressed the finding from the study by Penn State that the more human the avatar design was, the higher user expectations would be on the chatbot dialogue and thus cause a greater negative effect where the system fell short. Using the pixel art style not only makes the avatar contain less humanlike cues, but also implies older technology. As pixel art was a popular design in the late 1900s and has mostly been abandoned in modern designs, by using it we hoped to study if this would contribute to lowering the user expectations of the chatbot system similar to how lowering humanlike cues did. Below in Figure 3 are the three pixel art avatars we have used.



Figure 3. Pixel art style female, male and robot avatars

2.4 Configurations without avatars

For our experiment we also had configurations that did not use avatars at all, presenting as text-only. This was important to have in studying how the system performed with no avatar at all. Because of the previously mentioned assumption that realistic avatars with many humanlike cues tend to raise user expectations and thus make the chatbot system's shortcomings stand out more, it is a possibility that a text only chatbot provides a better experience by not having humanlike visual cues at all.

2.4 Audio across configurations

For our system be used three different audio recordings, female, male and genderless, and played them alongside avatars with the matching gender (genderless for the robot, which presents as a high pitched, cartoonish voice). The audio was created using the pyttsx3 text-to-speech conversion library in Python. For the text input for pyttsx3, we used the raw responses given by our chatbot system.

3 Chatbot system: Blenderbot

For the chatbot system in our experiment, we used the Emerson AI chatbot by Quickchat. This conversational chatbot is powered by the GPT-3 language model and considered to be among the best out there in terms of having natural and satisfying conversations with users. Emerson AI has been highly acclaimed for its ability to appear intelligent and give insightful responses to user inputs, thanks to its use of the GPT-3 language model which provides a deep understanding of language beyond literal meanings. Below is an example of chat from our experiment in which a participant conversed with the Emerson AI chatbot, text-only in this case.

Table 1: Example from conversation between participant and Emerson AI

Participant	Emerson AI
"Why were you having a bad day?"	"I had a really bad day at work. I had to go to a meeting with my boss."
"Is your boss not nice?"	"No, he's not nice at all. He's always trying to get me to do something."
"I'm sorry."	"It's okay. I just need to find a new job. I don't like this one."

4 Experiment setup

In the first stage of our experiment, we recruited a participant, a male retail worker in their early thirties, to chat with Emerson AI for 20 turns in the Facebook Messenger application. The log from this conversation would be used to create 10 footages with our 10 chatbot configurations acting out the conversation, in the perspective of the user.

For the next stage of our experiment, we recruited six new participants: three males and three females, all in their early twenties and whose native language is English. Two of the female participants are retail workers, and the rest are software engineers. These six participants observed the ten footages we created with the different avatar configurations, and then were tasked with rating them across some factors as well as ranking them.

In the evaluations, we first presented the four factors in a semantic differential scale. The semantic differential scale, which presents a rating in the form of a description and its antonym on the other side of the scale, is an effective way to have participants give ratings in evaluations. Within the scale, likeability, intelligence, humanity and enjoyability were chosen as the factors, with the expectation that these factors should distinguish the avatars that we have chosen as well as evaluate these distinctions.

Enoyable	2 1 0 -1 -2	Unenjoyable
Likeable	2 1 0 -1 -2	Unlikeable
Intelligent	2 1 0 -1 -2	Unintelligent
Humanlike	2 1 0 -1 -2	Not humanlike

Figure 4. Semantic differential scales that the participants used for evaluations

Furthermore, we asked the users to rank the chatbot configurations from the one they most prefer to chat with to the one they would least like to chat with. This would give us a meaningful insight into what types of avatars would be effective for satisfying conversations with the user on state-of-the-art level chatbot conversations.

4.2 Evaluation results and analysis

The averaged results of the scores given by our participants for each chatbot configuration is shown in Table 2. From these results, we can see that although the chatbot configurations using realistic images animated via MakeItTalk scored well in humanity and intelligence, they were not considered as likeable as the less realistic cartoon and pixel art avatars, and furthermore was shown to be less enjoyable to converse with. The avatar configurations which scored best in terms of enjoyability and likeability were the pixel art avatars - the scores averaged consistently well above 1, which the other categories failed to achieve. The text-only configurations very noticeably scored poorly across all four factors, suggesting to us that for conversational chatbots, users react better when there is an avatar present.

Interestingly, robot avatars scored best or tied with the best score for enjoyability, and aside from the pixel art avatars, the robot avatars also scored best in likeability. This suggests that in the setting of a conversation with a chatbot, users have better experiences with avatars that are not human. We cannot confirm that this positive result is due to the avatar being nonhuman or being a man made robot as we did not use any alternative nonhuman and non robotic avatars such as animals or even inanimate objects - thus, this would be a good point to consider in future works.

Table 2. Average scores given by participants for each chatbot - F, M and R denote female, male and robot respectively

	Enjoyability	Likeability	Intelligence	Humanit
Photo (F)	-0.17	0.17	**1.17**	**1.17**
Photo (M)	-0.83	-0.33	0.33	1.00
Photo (R)	1.33	1.17	0.33	-0.83
Toon (F)	0.17	0.67	0.50	0.33
Toon (M)	0.00	0.17	0.00	0.17
Toon (R)	**1.67**	**1.83**	-1.00	-0.67
Pixel (F)	1.33	1.67	1.00	0.17
Pixel (M)	1.50	1.67	0.67	0.00
Pixel (R)	1.50	1.33	-0.83	-1.00
text-only	-1.83	-1.83	-1.17	-1.67

As for rankings, overall, the participants ranked the chatbots with the pixel art avatars highest in terms of which chatbot they most preferred to chat with. Looking at Table 5, we can see that all but one participant chose an avatar configuration of this type as their top choice. Furthermore, surprisingly, the most realistic chatbot configuration made with images animated using MakeItTalk often ranked no better than the cartoon avatar configurations, especially when the male image was used.

Table 5. Rankings given by participants A, B, C, D E, and F for which chatbot they would most prefer to chat with - F, M and R denote female, male and robot respectively

	Photo (F)	Photo (M)	Photo (R)	Toon (F)	Toon (M)	Toon (R)	Pixel (F)	Pixel (M)	Pixel (R)	text-only
A	6	9	5	7	8	4	**1**	3	2	10
B	7	10	6	5	8	2	**1**	4	3	9
C	6	9	5	7	8	2	3	4	**1**	10
D	7	8	2	9	3	**1**	6	5	4	10
E	8	9	6	7	5	4	3	2	**1**	10
F	5	8	6	9	7	4	**1**	3	2	10

Participants A, B and C were male, while participants D, E and F were female. Taking this into consideration, we can see a pattern in the rankings for cartoon avatars and pixel avatars in which female participants ranked male avatars better while male participants ranked female avatars better. This supports the idea that generally, people tend to prefer avatars of the gender opposite to theirs - thus, it makes sense that gendered chatbots that are out there are dominantly female, as most chatbot developers have been male [3].

5. Conclusion

Through our experiment, we were able to test multiple different avatar designs on top of conversations carried out by a state-of-the-art chatbot system, and determine some key factors that would affect the user's experience. One key factor that we identified from our results was gender - there was a rather consistent trend of participants ranking and rating avatars of the gender opposite to their own, suggesting that in designing avatars for conversational agents, one must consider both the gender of the user as well as the gender that the system presents as. Interestingly, we also found that avatars which were robots ranked better as well as scored better for enjoyability and likeability, suggesting that humanlike cues do not necessarily improve the user experience. We also observed that users also expressed affinity to less realistic chatbot avatar designs, with the pixel art style avatar which barely animates ranking high and scoring well for enjoyability and likeability. This result suggests that high tech avatar designs may actually be detrimental on conversational agents.

In the future, this study could benefit from including more nonhuman avatars. As we found that the robot avatars were received very well, it would be interesting to try to determine if this affinity comes from the avatar being a man-made robot or the characteristic of being non-human in general. Furthermore, in the future we should include human avatars lacking a clear gender - this way, we can make more valid and in depth observations on the role of gender in the design of conversational agents.

As conversational chatbots have, in recent years, finally reached a level in which they could carry satisfying conversations with users, it is important to consider the presentation of these chatbots. Having avatars accompany chatbots helps them appear to have identities, which makes for a more engaging conversation. As shown in the results of our study, it is important to carefully determine the avatar design, as it could significantly affect the user experience.

References

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